Important information. Please read carefully.

Adroten™

BISOPROLOL FUMARATE 5MG
Film-Coated Tablet

COMPOSITION
Each film-coated tablet contains 5mg Bisoprolol Fumarate.

PHARMACODYNAMICS
Bisoprolol is a cardioselective beta blocker. It is a potent and highly $beta_1$-selective adrenoreceptor blocking agent that devoid of intrinsic sympathomimetic and membrane stabilizing activity.

PHARMACOKINETICS
Bisoprolol is almost completely absorbed from the gastrointestinal tract and undergoes only minimal first-pass metabolism resulting in an oral bioavailability of about 90%.

Peak plasma concentrations are reached within 2 to 4 hours after oral doses. The plasma protein binding of bisoprolol is about 30%.

Bisoprolol is moderately lipid-soluble. It is metabolized in the liver and excreted in urine, about 50% as unchanged drug and 50% as metabolites. It has a plasma elimination half-life of 10 to 12 hours.

INDICATIONS
- Treatment of high blood pressure (hypertension).
- Treatment of coronary heart disease (angina pectoris).
- Treatment of stable chronic heart failure with reduced systolic left ventricular function in addition to ACE inhibitors, and diuretics, and optionally cardiac glycosides.

DOSAGE AND ADMINISTRATION
Adroten Film-Coated Tablet 5mg should be taken in the morning, with or without food. Do not chew the tablet and swallow it with liquid.

Treatment of hypertension or angina pectoris
Treatment of hypertension or angina pectoris. In all cases the dose regimen is adjusted individually by your doctor, in particular according to the pulse rate and therapeutic success.

The usual dose of bisoprolol is 5 to 10 mg orally as a single daily dose. The maximum recommended dose is 20 mg daily. A reduction in dose may be necessary in patients with hepatic or renal impairment.

Adroten Film-Coated Tablet 5mg must be used with caution in patients with hypertension or angina pectoris and accompanying heart failure.

Treatment of stable chronic heart failure
The initiation of treatment of stable chronic heart failure with Adroten Film-Coated Tablet 5mg necessitates a special titration phase.

Precondition for treatment with bisoprolol is stable chronic heart failure without acute failure.

The treatment of stable chronic heart failure with bisoprolol is initiated according to the following titration scheme, individual adaptation may be necessary depending on how well the patient tolerates each dose, i.e. the dose is to be increased only, if the previous dose is well tolerated.

1st week: 1.25 mg bisoprolol fumarate once daily.

2nd week: 2.5 mg bisoprolol fumarate (½ tablet Adroten Film-Coated Tablet 5mg) once daily.

3rd week: 3.75 mg bisoprolol fumarate once daily.

4th - 7th week: 5mg bisoprolol fumarate (1 tablet Adroten Film-Coated Tablet 5mg) once daily.

8th - 11th week: 7.5mg bisoprolol fumarate (1 tablet Adroten Film-Coated Tablet 5mg) once daily.

12th week and beyond: 10 mg bisoprolol fumarate (2 tablets Adroten Film-Coated Tablet 5mg)

The maximum recommended dose is 10 mg bisoprolol fumarate once daily.

Close monitoring of vital signs (blood pressure, heart rate) and symptoms of worsening heart failure is recommended during the titration phase. Symptoms may already occur within the first day after initiating therapy.

Treatment modification : If during the titration phase or thereafter, transient worsening of heart failure, hypotension or bradycardia occurs, reconsideration of the dosage of concomitant medication is recommended. It may also be necessary to temporarily lower the dose of bisoprolol or to consider discontinuation. The reintroduction and/or uptitration of bisoprolol should always be considered when the patient becomes stable again.

Duration of treatment: Treatment with Adroten Film-Coated Tablet 5mg is generally a long-term therapy. Do not stop treatment abruptly or change the recommended dose without talking to your doctor first since this might lead to a transitory worsening of heart condition. Especially in patients with ischaemic heart disease, treatment must not be discontinued suddenly. If discontinuation is necessary, the daily dose is gradually decreased.

Special populations
- Renal or hepatic impairment
  In hypertension and angina pectoris patients with severe hepatic impairment or with a creatinine clearance of less than 20mL/minute, a maximum daily dose of 10 mg bisoprolol is recommended.

  In patients with liver or kidney function disorders of mild to moderate severity no dosage adjustment is normally required.

  There is no information regarding pharmacokinetics of bisoprolol in patients with chronic heart failure and concomitant hepatic or renal impairment. Titration of the dose in these populations must therefore be made with particular caution.
- Elderly
  No dosage adjustment is required.
- Children
  No pediatric experience with bisoprolol, therefore, its use cannot be recommended for children.

CONTRAINDICATIONS
- Acute heart failure or during episodes of heart failure decompensation requiring intravenous therapy with substances increasing the contractility of the heart.
- Shock induced by disorders of cardiac function (cardiogenic shock).
- Severe disturbances of atrioventricular conduction (second or third degree AV block) without a pacemaker.
- Sick sinus syndrome.
- Sinoatrial block.
- Bradycardia with heart beat less than 60 beats/min.
- Hypotension with systolic blood pressure less than 100 mm Hg.
- Severe bronchial asthma or severe chronic obstructive pulmonary disease.
- Severe forms of peripheral arterial occlusive disease or Raynaud's syndrome
- Untreated tumors of the adrenal gland (phaeochromocytoma).
- Metabolic acidosis.
- Hypersensitivity to bisoprolol or to any of the excipients.

WARNING AND PRECAUTIONS
- Diabetes mellitus with extremely fluctuating blood glucose levels where symptoms of hypoglycemia such as tachycardia, palpitations or sweating can be masked.
- Strict fasting.
- Ongoing desensitization therapy.
- Mild disturbances of atrioventricular conduction (first degree AV block).
- Disturbed blood flow in the coronary vessels due to vasospasms (Prinzmetal's angina).
- Peripheral arterial occlusive disease (intensification of complaints may occur especially when starting therapy).
- Patients with a personal or family history of psoriasis.
- *Respiratory system:* In bronchial asthma or other symptomatic chronic obstructive pulmonary diseases concomitant bronchodilator therapy is indicated. An increase in airway resistance may occasionally occur in patients with asthma, requiring a higher dose of $beta_2$-sympathomimetics.
- *Allergic reactions:* Beta-blocker, including bisoprolol tablets, may increase the sensitivity to allergens and the severity of anaphylactic reactions because the adrenergic counter regulation under beta-blockade may be alleviated. Treatment with adrenaline may not always give the expected therapeutic effect.
- *General anesthesia:* In patients undergoing general anesthesia the anesthetist must be aware of beta-blockade. If it is thought necessary to withdraw bisoprolol tablets before surgery, this should be done gradually and completed about 48 hours prior to anesthesia.
- *Phaeochromocytoma:* In patients with a tumor of the adrenal gland, bisoprolol tablets may only be administered after previous alpha-receptor blockade.
- *Thyrotoxicosis:* Under treatment with bisoprolol tablets, the symptoms of a thyroid hyper function may be masked.

DRUG INTERACTIONS
Combinations not recommended when used concomitantly with bisoprolol tablets
- Treatment of stable chronic heart failure Class-I antiarrhythmic medicines (e.g. quinidine, disopyramide, lidocaine, phenytoin; flecainide, propafenone) may increase the depressant effect of Adroten Film-Coated Tablet 5mg on atrio-ventricluar impulse conduction and the contractility of the heart.
- Calcium antagonists of the verapamil type and to a lesser extent of the diltiazem type may lead to reduced contractility of the heart muscle and delayed atrio-ventricular impulse conduction. Especially intravenous administration of verapamil in patients on beta-blocker treatment may lead to profound hypotension and atrioventricular block.
- Centrally acting blood pressure-lowering medicines (such as clonidine, methyldopa, moxonodine, rilmenidine) may lead to a reduction of heart rate and cardiac output, as well as to vasodilatation due to a decrease in the central sympathetic tonus. Abrupt withdrawal, particularly if prior to beta-blocker discontinuation, may increase risk of "rebound hypertension".

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Combinations to be used with cautions when used concomitantly with bisoprolol tablets
- Calcium antagonists of the dihydropyridine type (e.g. nifedipine) may increase the risk of hypotension. An increased risk of a further deterioration of the ventricular pump function in patients with heart failure cannot be excluded.
- Class-III antiarrhythmic medicines (e.g. amiodarone) may increase the inhibitory effect of bisoprolol tablet on atrio-ventricular impulse conduction.
- Topical beta-blocker (e.g. eye drops for glaucoma treatment) may add to the systemic effects of bisoprolol tablets.
- Parasympathomimetic medicines may increase the inhibitory effect on atrio-ventricular impulse conduction and the risk of bradycardia.
- The blood sugar lowering effect of insulin or oral antidiabetic medicines may be intensified. Warning signs of hypoglycemia, especially tachycardia may be masked or suppressed. Such interactions are considered to be more likely with nonselective beta-blockers.
- Anesthetic agents may increase the risk of cardio depressive actions of bisoprolol tablets, leading to hypotension.
- Cardiac glycosides (digitalis) may lead to an increase in impulse conduction time and thus reduction in heart rate.
- Non-steroidal anti-inflammatory (NSAIDs) may reduce the blood pressure lowering effect of bisoprolol tablets.
- Beta-sympathomimetics (e.g. isoprenaline, dobutamine) may lead to a reduced effect of both agents.
- Sympathomimetics that activate both beta and alpha-adrenoceptors (e.g. noradrenaline, adrenaline) may intensify the alpha-adrenoceptor-mediated vasoconstrictor effects of these agents leading to blood pressure increase and exacerbated intermittent claudication. Such interactions are considered to be more likely with nonselective beta-blockers.
- Antihypertensive agents as well as other medicines with blood pressure lowering potential (e.g. tricylic antidepressants, barbiturates, pheothiazines) may increase the blood pressure lowering effect of bisoprolol tablets.

Combinations to be considered when used concomitantly with bisoprolol tablets
- Mefloquine may increase the risk of decelerating the heart rate (bradycardia).
- Monoamine oxidase inhibitors (except MAO-B inhibitors) may enhance the hypotensive effect of the beta-blockers, and may also be a risk for hypertensive crisis.

PREGNANCY AND LACTATION
Pregnancy
Beta-blockers reduce placental blood flow and may affect the development of the unborn child, thus only recommended with careful assessment of benefit-to-risk ratio by the doctor.

Placental and uterine blood flow as well as the growth of the unborn child must be monitored and, in case of harmful effects on pregnancy or the fetus, alternative therapeutic measures considered. The newborn infant must be monitored closely after delivery. Symptoms of reduced blood glucose and slowed pulse rate generally may occur within the first 3 days of life.

Lactation
It is not known whether this drug is excreted in human milk. Therefore, breastfeeding is not recommended during administration of bisoprolol.

SIDE EFFECTS
Below are the adverse reactions that might occur after taken bisoprolol.

The frequency is defined as follows:

Very Common (≥1/10)

Common (≥1/100 to <1/10)

Uncommon (≥1/1,000 to <1/100)

Rare (≥1/10,000 to <1/1,000)

Very Rare (<1/10,000).

- Cardiac disorders
  Very common: bradycardia
  Common: worsening of heart failure
  Uncommon: AV-conduction disturbances
- Nervous system disorders
  Common: dizziness, headache
  Rare: syncope
- Eye disorders
  Rare: reduced tear flow (to be considered if the patient uses lenses)
  Very rare: conjunctivitis
- Ear and labyrinth disorders
  Rare: hearing impairment
- Respiratory, thoracic and mediastinal disorders
  Uncommon: bronchospasm in patients with bronchial asthma or a history of obstructive airways disease.
  Rare: allergic rhinitis
- Gastrointestinal disorders
  Common: gastrointestinal complaints such as nausea, vomiting, diarrhea, constipation
- Skin and subcutaneous tissue disorders
  Rare: hypersensitivity reactions (itching, flush, rash)
  Very rare: beta-blockers may provoke or worsen psoriasis or induce psoriasis-like rash, alopecia
- Musculoskeletal and connective tissue disorders
  Uncommon: muscular weakness and cramps
- Vascular disorders
  Common: feeling of coldness or numbness in the extremities, hypotension
  Uncommon: orthostatic hypotension
- General disorders
  Common: asthenia, fatigue
- Hepatobiliary disorders
  Rare: hepatitis
- Reproductive system and breast disorders
  Rare: potency disorders
- Psychiatric disorders
  Uncommon: sleep disorders, depression
  Rare: nightmares, hallucinations

SYMPTOMS AND TREATMENT FOR OVERDOSAGE
The most common signs with overdosage of a beta-blocker are bradycardia, hypotension, bronchospasm, acute cardiac insufficiency and hypoglycemia. Dose of 2000 mg bisoprolol is considered as overdose. If overdose occur, bisoprolol treatment should be stopped and supportive and symptomatic treatment should be provided.

Bradycardia
Administer intravenous atropine. If the response is inadequate, isoprenaline or another agent with positive chronotropic properties may be given cautiously. Under some circumstances, transvenous pacemaker insertion may be necessary.

Hypotension
Intravenous fluids and vasopressors should be administered. Intravenous glucagon may be useful.

AV block (second or third degree)
Patients should be carefully monitored and treated with isoprenaline infusion or transvenous cardiac pacemaker insertion.

Acute worsening of heart failure
Administer intravenous diuretics, inotropic agents, vasodilating agents.

Bronchospasm
Administer bronchodilator therapy such as isoprenaline, $beta_2$-sympathomimetic drugs and/or aminophylline.

Hypoglycemia
Administer intravenous glucose.

STORAGE CONDITION
Store below 30°C.

SHELF-LIFE
The expiry date is indicated on the packaging.

PRODUCT DESCRIPTION
Yellowish white, round, normal convex film-coated tablet, with "XS" marking on one side and plain on reverse.

Available as blister strips of 10's in packing of 100 tablets per box.

KEEP OUT OF REACH OF CHILDREN / JAUHI DARI KANAK-KANAK
For further information, please consult your pharmacist or physician.

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Manufactured and Marketed by:
Xepa-Soul Pattinson (Malaysia) Sdn Bhd
1-5, Cheng Industrial Estate, 75250 Melaka, Malaysia.

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